

Homeostasis — Complete Notes

Grade 12

Life Sciences

CAPS

What you'll learn

- what homeostasis is and why it matters
- the general structure of a negative feedback loop
- how the body controls temperature (thermoregulation)
- how the kidneys control water and salt balance (osmoregulation)
- how the body controls blood glucose using insulin and glucagon
- what happens when homeostasis fails, such as in diabetes mellitus
- how to interpret graphs and data in homeostasis questions

What is homeostasis?

Definition — homeostasis: the maintenance of a constant internal environment within narrow limits, despite changes in the external environment. The internal environment is the tissue fluid that surrounds body cells. Keeping its temperature, water content, salt concentration, pH and glucose concentration stable allows enzymes and cells to work at their best.

Humans are homeotherms (endotherms): we keep our core body temperature almost constant at about 37 °C, whatever the weather.

The negative feedback loop

Most homeostatic control uses negative feedback. Negative here means the response reverses (opposes) the original change. A negative feedback loop has the following parts:

1. **Stimulus** — a change away from normal, for example the body temperature rises.
2. **Receptor** — a structure that detects the change, for example thermoreceptors in the skin and hypothalamus.
3. **Control centre** — the part that processes the information and decides the response, usually the hypothalamus, pituitary or pancreas.
4. **Effector** — the organ or gland that carries out the response, for example skin blood vessels, sweat glands, the kidneys or the liver.
5. **Response** — the action taken, for example sweating increases.
6. **Result** — the original change is reversed and the internal environment returns to normal.

If the response makes a change bigger instead of reversing it, this is positive feedback (for example, the release of oxytocin during childbirth).

```
NORMAL (set point)
|
stimulus: a change occurs
v
receptor detects the change
v
control centre processes the information
v
effector responds
v
response reverses the change ---> back to NORMAL
```

Thermoregulation

Definition — thermoregulation: the control of body temperature to keep it close to 37 °C. The hypothalamus is the control centre and receives messages from thermoreceptors in the skin and in the hypothalamus itself.

When the body becomes too hot

Stimulus: the core temperature rises above 37 °C, for example during exercise or on a hot day.

- Blood vessels in the skin vasodilate (widen), so more blood flows near the skin surface and more heat is lost by radiation and convection.
- Sweat glands produce more sweat; as the sweat evaporates it takes heat from the skin and cools the body.
- The hair erector muscles relax, so the hairs lie flat and less air is trapped, reducing insulation.
- Shivering stops and the metabolic rate is lowered, so less heat is produced.

Result: the body temperature falls back to about 37 °C.

When the body becomes too cold

Stimulus: the core temperature falls below 37 °C, for example on a cold day.

- Blood vessels in the skin vasoconstrict (narrow), so less blood flows near the skin surface and less heat is lost.
- Sweat glands produce little or no sweat, reducing heat loss by evaporation.
- Shivering starts: rapid involuntary muscle contractions release heat as a by-product of respiration.
- The hair erector muscles contract, so the hairs stand upright and trap a layer of air that insulates the body.
- The metabolic rate increases (more thyroxine and adrenaline), which produces more heat.

Result: the body temperature rises back to about 37 °C.

Condition	Skin blood vessels	Sweating	Hair muscles	Shivering
Too hot	Vasodilation	Increases	Relax (hair flat)	Reduced
Too cold	Vasoconstriction	Decreases	Contract (hair up)	Increases

Osmoregulation (the kidneys)

Definition — osmoregulation: the control of the water and salt balance of the body fluids, keeping the salt concentration and water potential within narrow limits. The kidneys are the main effectors.

The kidney and the nephron

Each kidney contains about a million tiny functional units called nephrons. Each nephron includes the following parts:

- Bowman's capsule, which surrounds a knot of capillaries called the glomerulus. High pressure forces water, salts, glucose and urea out of the blood into the capsule. This is ultrafiltration. Proteins and blood cells are too large to pass through, so they stay in the blood.
- Proximal convoluted tubule, where useful substances (glucose, amino acids, most salts and much water) are selectively reabsorbed back into the blood.
- Loop of Henle, which helps the kidney produce urine of different concentrations.
- Distal convoluted tubule and collecting duct, where the amount of water reabsorbed is adjusted by the hormone ADH.

Definitions: ultrafiltration is the filtering of blood under pressure in the glomerulus and Bowman's capsule. Selective reabsorption is the movement of useful substances from the nephron back into the blood.

```
Blood in ---> [ glomerulus inside Bowman's capsule ]   ultrafiltration
                |
                v
proximal tubule --> loop of Henle --> distal tubule --> collecting duct
(reabsorb glucose,      (helps concentrate   (adjust water      (ADH controls
 amino acids, salts,    the urine)           reabsorbed)      water reabsorbed)
 most water)

                |
                v
                urine ---> ureter ---> bladder ---> urethra
```

ADH and water balance

ADH (antidiuretic hormone) is produced by the hypothalamus and released by the posterior pituitary gland. It makes the walls of the collecting duct more permeable to water.

If the body is short of water (dehydration, salty food, heavy sweating):

- The water potential of the blood falls, so the blood is too concentrated.
- Osmoreceptors in the hypothalamus detect this.
- The posterior pituitary releases more ADH.
- The collecting duct becomes more permeable and reabsorbs more water.
- A small volume of concentrated urine is produced, and the water content of the blood rises back toward normal.

If the body has too much water (after drinking a lot of water):

- The water potential of the blood rises, so the blood is too dilute.
- Osmoreceptors in the hypothalamus detect this.
- Less ADH is released.
- The collecting duct becomes less permeable and reabsorbs less water.

- A large volume of dilute urine is produced, and the water content of the blood falls back toward normal.

Blood water potential	ADH released	Water reabsorbed in collecting duct	Volume of urine	Concentration of urine
Low (too little water)	More	More	Small	Concentrated
High (too much water)	Less	Less	Large	Dilute

Blood glucose regulation

The normal blood glucose concentration is about 4 to 7 mmol/L. It is controlled by two antagonistic hormones made in the islets of Langerhans in the pancreas: insulin (from the beta cells) lowers blood glucose, and glucagon (from the alpha cells) raises it.

When blood glucose is too high (after a meal)

Stimulus: glucose is absorbed from the small intestine, so blood glucose rises.

- The beta cells of the pancreas detect the rise and secrete insulin.
- Insulin travels in the blood to the liver and to body cells.
- The liver converts glucose to glycogen and stores it.
- Body cells absorb more glucose and use it in respiration.

Result: blood glucose falls back to normal.

When blood glucose is too low (fasting or exercise)

Stimulus: cells use up glucose in respiration, so blood glucose falls.

- The alpha cells of the pancreas detect the fall and secrete glucagon.
- Glucagon travels in the blood to the liver.
- The liver converts stored glycogen back to glucose and releases it into the blood.

Result: blood glucose rises back to normal.

Hormone	Made by	Main effect	Net result
Insulin	Beta cells of the pancreas	Converts glucose to glycogen in the liver and increases glucose uptake by cells	Lowers blood glucose
Glucagon	Alpha cells of the pancreas	Converts glycogen to glucose in the liver	Raises blood glucose

When homeostasis fails — diabetes mellitus

Definition — diabetes mellitus: a condition in which blood glucose stays too high (hyperglycaemia) because insulin is not made or does not work properly.

- Type 1 diabetes: the beta cells are damaged or destroyed, so too little or no insulin is produced. It is treated with insulin injections.
- Type 2 diabetes: the body still makes insulin but the cells respond poorly to it (insulin resistance). It is linked to obesity, a poor diet and lack of exercise, and is managed mainly with lifestyle changes and

sometimes medication.

Common symptoms include high blood glucose, glucose in the urine, frequent urination, thirst, tiredness and weight loss. Untreated diabetes can damage the kidneys, eyes, nerves and blood vessels.

Worked examples

Worked example 1 — working through a negative feedback loop

Problem: On a very hot day a runner's body temperature rises. Describe, step by step, how the body returns the temperature to normal.

Step 1 — identify the stimulus. The core body temperature rises above the set point of 37 °C. Step 2 — identify the receptor. Thermoreceptors in the skin and in the hypothalamus detect the increase. Step 3 — identify the control centre. The hypothalamus processes the information. Step 4 — identify the effectors and their responses.

- Skin arterioles vasodilate, so more blood flows near the surface and more heat is lost by radiation and convection.
- Sweat glands secrete more sweat, and the evaporation of the sweat removes heat from the skin.
- The hair erector muscles relax so the hairs lie flat, reducing insulation.
- Shivering stops and the metabolic rate drops, so less heat is produced.

Step 5 — state the result. The body temperature is lowered back toward 37 °C, so the original change is reversed. This is negative feedback.

Answer summary: receptor (thermoreceptors) to control centre (hypothalamus) to effectors (skin blood vessels, sweat glands, hair muscles) to response (vasodilation, sweating, flattened hair) to a fall in temperature.

Worked example 2 — reading a blood glucose graph

Problem: A person drinks a glucose drink at time 0. The blood glucose concentration is measured every 30 minutes.

Time after drinking (minutes)	Blood glucose (mmol/L)
0	5.0
30	8.5
60	7.0
90	5.6
120	5.1

(a) Describe the trend of blood glucose over the 120 minutes. (b) Explain what causes the rise and the later fall.

Step 1 — read the data. Glucose rises from 5.0 to a peak of 8.5 mmol/L at 30 minutes, then falls back to about 5.1 mmol/L by 120 minutes. Step 2 — explain the rise. The glucose drink is absorbed from the small intestine into the blood, so the blood glucose concentration increases. Step 3 — explain the fall. The pancreas detects the high blood glucose, and the beta cells secrete insulin. Insulin causes the liver to convert glucose to glycogen and body cells to take up glucose, so blood glucose decreases. Step 4 — link to homeostasis.

Because the response (insulin) reverses the original rise, this is negative feedback, and the value returns close to the normal range of 4 to 7 mmol/L.

Worked example 3 — predicting the effect of ADH

Problem: A person does not drink any water for many hours on a hot day. Predict the effect on ADH, on water reabsorption, and on the concentration and volume of urine. Explain your reasoning.

Step 1 — identify the change. The person loses water by sweating and does not replace it, so the blood becomes more concentrated and its water potential falls. Step 2 — identify the receptor and control centre. Osmoreceptors in the hypothalamus detect the fall in water potential. Step 3 — identify the response. The posterior pituitary releases MORE ADH, which makes the collecting duct more permeable, so MORE water is reabsorbed into the blood. Step 4 — predict the urine. A small volume of concentrated urine is produced. Step 5 — explain the benefit. Reabsorbing more water protects the body from dehydration and reverses the change in the blood, so this is negative feedback.

Answer: ADH increases; water reabsorption increases; urine volume decreases; urine concentration increases.

Common mistakes

- Describing only the response. Always give the full loop with the stimulus and receptor as well.
- Getting vasodilation and vasoconstriction the wrong way round. Vasodilation widens the vessels so more heat is lost; vasoconstriction narrows them so less heat is lost.
- Confusing insulin and glucagon. Insulin lowers blood glucose; glucagon raises it.
- Writing that insulin cures diabetes. An insulin injection treats Type 1 diabetes but does not cure it.
- Confusing osmoregulation (water and salt balance) with thermoregulation (temperature).
- Saying ADH increases urine volume. More ADH means more water is reabsorbed, so less urine is produced.
- Forgetting that ADH is released by the posterior pituitary, not by the kidney.
- Mixing up ultrafiltration (in the Bowman's capsule) with selective reabsorption (in the tubule).
- Ignoring units and set points in data questions (blood glucose in mmol/L, temperature in °C).

Quick summary / cheat sheet

- Homeostasis keeps the internal environment constant within narrow limits.
- Negative feedback: stimulus, receptor, control centre, effector, response, then back to normal.
- Too hot: vasodilation, sweating, hair flat, less shivering.
- Too cold: vasoconstriction, little sweating, shivering, hair up, higher metabolic rate.
- Nephron order: Bowman's capsule (ultrafiltration), proximal tubule, loop of Henle, distal tubule, collecting duct (ADH).
- Too little water: more ADH, more water reabsorbed, small volume of concentrated urine.
- Too much water: less ADH, less water reabsorbed, large volume of dilute urine.
- High blood glucose: insulin from the beta cells turns glucose into glycogen, so glucose falls.
- Low blood glucose: glucagon from the alpha cells turns glycogen into glucose, so glucose rises.
- Diabetes mellitus is too high blood glucose; Type 1 means little or no insulin, Type 2 means insulin resistance.

Practice questions

1. Define homeostasis.
2. List the five main parts of a negative feedback loop.
3. Name the control centre for thermoregulation in the human body.
4. Explain how sweating helps to cool the body on a hot day.
5. A person drinks a large volume of water. Describe the effect on ADH, on water reabsorption and on the volume of urine.
6. Name the hormone that lowers blood glucose and the cells in the pancreas that secrete it.
7. Distinguish between Type 1 and Type 2 diabetes mellitus.
8. Explain why the water content of the blood stays fairly constant even when a person sweats a lot.

Answers

1. Homeostasis is the maintenance of a constant internal environment within narrow limits, despite changes in the external environment.
2. Stimulus, receptor, control centre, effector and response (with the result being that the change is reversed).
3. The hypothalamus.
4. Sweat is released onto the skin by the sweat glands; as the water in the sweat evaporates it takes heat from the skin, so the skin and the blood flowing through it cool down.
5. The blood becomes more dilute, so its water potential rises. Osmoreceptors detect this, the posterior pituitary releases LESS ADH, the collecting duct reabsorbs LESS water, and a LARGE volume of dilute urine is produced.
6. Insulin, secreted by the beta cells of the islets of Langerhans in the pancreas.
7. In Type 1 diabetes the beta cells are damaged so little or no insulin is produced and it is treated with insulin injections; in Type 2 diabetes insulin is still produced but the body cells respond poorly to it (insulin resistance), so it is managed mainly with lifestyle changes and sometimes medication.
8. When a person sweats, the blood becomes more concentrated. Osmoreceptors in the hypothalamus detect this and cause the posterior pituitary to release more ADH, so the kidneys reabsorb more water and produce less urine, which returns the water content of the blood toward normal. This is negative feedback.

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